



Getting Started with Programming

The main things you need to
understand when getting started!

Defining Computer Science

Computer science is the systematic study of how problems can be solved using *computation*.

It involves analyzing problems, designing logical solutions, and expressing those solutions in a form a computer can execute.

The field includes topics such as **algorithms**, **data**, **abstraction**, and **automation**, not just programming languages.

Computer science emphasizes understanding *how and why solutions work*, rather than only producing outputs.

Computation is the process of transforming *input data* into *output results* by executing a specific, well-defined set of rules.

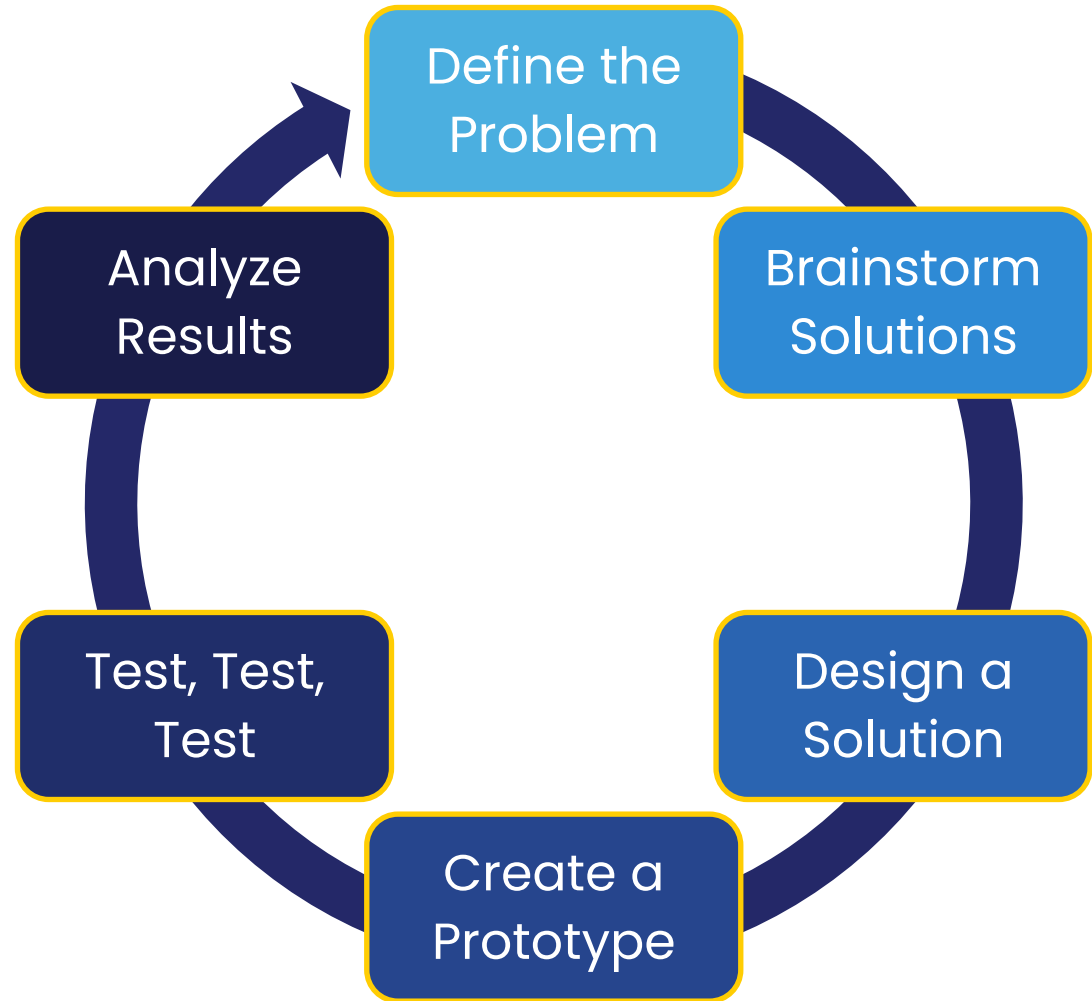
Problem Solving Skills

This course is designed to strengthen **problem-solving skills**, not just teach syntax.

You will learn to break complex problems into smaller, manageable steps using *logical reasoning*.

Emphasis is placed on understanding the problem before attempting to write code.

These skills are transferable and useful beyond computer science, including in math, science, and everyday decision-making.



Coding vs Computer Science



Coding is the act
of writing
instructions in a
programming
language.

Computer science
is the process of
developing the
solution that the
code represents.

A strong solution can be expressed in many languages, but weak logic cannot be fixed by syntax alone.

This distinction highlights why understanding the problem is more important than memorizing commands.

Why Learn Programming?

Programming is a **useful skill**, not just a career—you don't need to become a professional programmer to benefit from it.

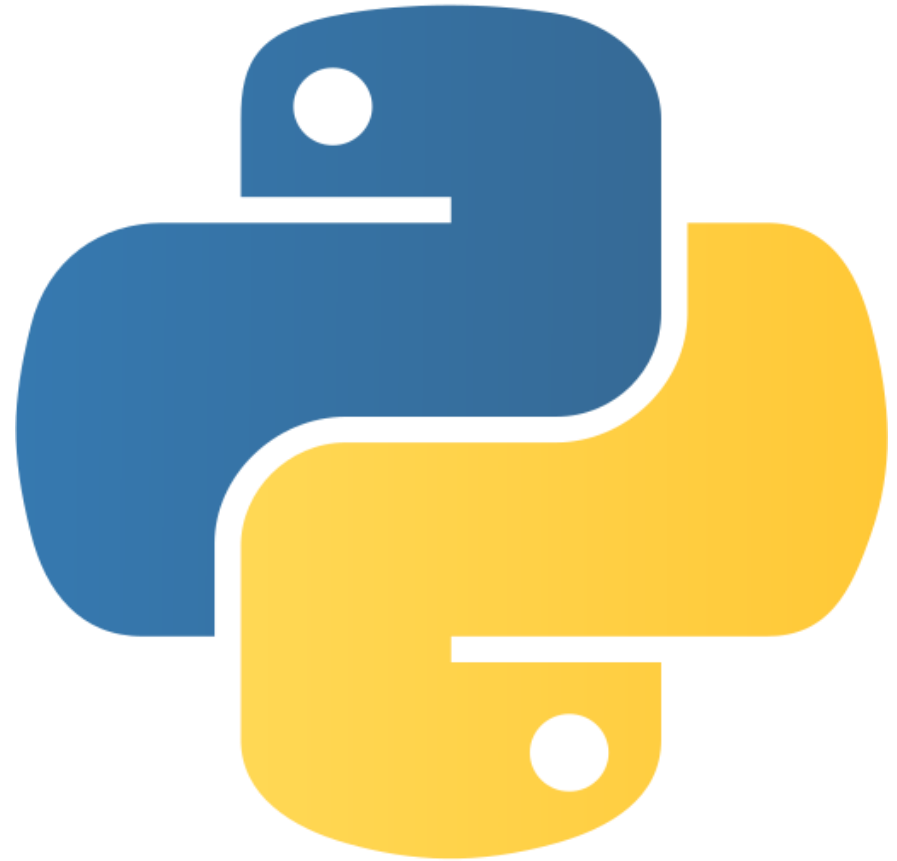
Automate everyday tasks that would otherwise be repetitive or time-consuming.

Create custom solutions designed around exactly what you need.

Become more **independent and efficient** instead of relying on existing software or other people.

Develop **critical thinking and problem-solving** by deciding how to approach problems.

Programming is also **creative**—there are often many different ways to solve the same problem.



Some Basics Before We Get Started

Before we get started, some basic concepts that you might already be familiar with:

- **Variables**
- **True/False statements**
- **Connected true and false statements**
- **Making decisions**
- **Flowcharts**